Lecture 2

Why quadratic surds have repeating continued fractions

First: the converse of Lagrange's theorem.

Then: when the repetition starts at the first digit.

The converse we still need

Lecture 1 proved

$$\text{eventually periodic CF} \implies \text{quadratic irrational.}$$

Today we prove

$$\text{quadratic irrational} \implies \text{eventually periodic CF.}$$

Every quadratic surd has an expansion of the form

$$[a_0; \ldots, a_{r-1}, \overline{b_0, \ldots, b_{\ell-1}}].$$

The finite prefix is allowed to be empty.

The algorithm and its remaining tails

Start with an irrational $x_0 = x$ and repeat

$$a_n = \lfloor x_n \rfloor, \qquad x_{n+1} = \frac{1}{x_n - a_n}.$$

Since $0 < x_n - a_n < 1$, we have $x_{n+1} > 1$.

$$x_n = a_n + \frac{1}{x_{n+1}} = [a_n; a_{n+1}, \ldots].$$

The complete quotient x_n is the remaining tail.

For $n \geq 1$, $a_n \geq 1$; the first digit a_0 can be any integer.

First see repetition happen: $\sqrt{3}$

Here $x_0 = \sqrt{3}$ and $a_0 = 1$. Rationalizing gives

$$x_1 = \frac{1}{\sqrt{3}-1} = \frac{\sqrt{3}+1}{2}, \qquad a_1 = 1,$$

$$x_2 = \frac{1}{(\sqrt{3}+1)/2 - 1} = \sqrt{3}+1, \quad a_2 = 2,$$

$$x_3 = \frac{1}{\sqrt{3}+1-2} = \frac{\sqrt{3}+1}{2} = x_1.$$

The same tail gives the same next digit and the same next tail.

$$\sqrt{3} = [1; \overline{1, 2}].$$

What will force a tail to return?

We will write every tail using the same square root:

$$x_n = \frac{P_n + \sqrt{D}}{Q_n}, \qquad P_n, Q_n \in \mathbb{Z}.$$

1. Find an exact rule for the two integers.
2. Prove that, eventually, both integers are bounded.
3. Apply the pigeonhole principle: a pair must repeat.

The conjugate will supply the bounds in Step 2.

A surd and its other root

Write $x = u + v\sqrt{D}$, where $u, v \in \mathbb{Q}$, $v \neq 0$, and D is a positive nonsquare integer.

The factorization

$$\begin{aligned}(t - (u + v\sqrt{D}))(t - (u - v\sqrt{D})) \\ = (t-u)^2 - v^2 D = t^2 - 2ut + u^2 - v^2 D\end{aligned}$$

has rational coefficients and two distinct roots.

Thus the other root, called the conjugate, is

$$x' = u - v\sqrt{D}.$$

Why conjugation respects arithmetic

Let $z = u + v\sqrt{D}$ and $w = r + s\sqrt{D}$, with rational coefficients.

$$(z + w)' = (u + r) - (v + s)\sqrt{D} = z' + w'.$$

For multiplication, expand both products:

$$zw = (ur + vsD) + (us + vr)\sqrt{D},$$
$$z'w' = (ur + vsD) - (us + vr)\sqrt{D} = (zw)'.$$

Rational numbers stay fixed: $c' = c$ for $c \in \mathbb{Q}$.

In particular, $(z - a)' = z' - a$ for an integer a.

Why conjugation respects reciprocals

For $z = u + v\sqrt{D} \neq 0$, rationalize:

$$\frac{1}{z} = \frac{u - v\sqrt{D}}{u^2 - v^2 D}.$$

Since $\sqrt{D}$ is irrational, $z \neq 0$ also gives $z' \neq 0$. Thus $zz' = u^2 - v^2 D$ is a nonzero rational number.

Now change the sign of the square root:

$$\left(\frac{1}{z}\right)' = \frac{u + v\sqrt{D}}{u^2 - v^2 D} = \frac{1}{z'}.$$

For $a, b, c, d \in \mathbb{Q}$ and $cz + d \neq 0$, this gives

$$\left(\frac{az + b}{cz + d}\right)' = \frac{az' + b}{cz' + d}.$$

Step 1: start with an integer pair

Take $Ax^2 + Bx + C = 0$ with $A, B, C \in \mathbb{Z}$, $A \neq 0$.

Choose the common sign of A, B, C so that

$$x = \frac{-B + \sqrt{D}}{2A}, \qquad D = B^2 - 4AC.$$

Then

$$P_0 = -B, \quad Q_0 = 2A, \quad x = \frac{P_0 + \sqrt{D}}{Q_0}.$$

Moreover,

$$D - P_0^2 = -4AC = Q_0(-2C), \qquad Q_0 \mid D - P_0^2.$$

D is positive and nonsquare; Q_0 may be negative.

Derive the rule by rationalizing

Suppose $x_n = (P_n + \sqrt{D})/Q_n$ and $a_n = \lfloor x_n \rfloor$. Set

$$P_{n+1} = a_n Q_n - P_n.$$

Then

$$\begin{aligned} x_{n+1} &= \frac{Q_n}{\sqrt{D} - P_{n+1}} \\ &= \frac{Q_n(\sqrt{D} + P_{n+1})}{D - P_{n+1}^2} = \frac{P_{n+1} + \sqrt{D}}{Q_{n+1}}, \end{aligned}$$

where

$$Q_{n+1} = \frac{D - P_{n+1}^2}{Q_n}.$$

Why the new pair is still integral

Assume $Q_n \mid D - P_n^2$. Since

$$P_{n+1}^2 - P_n^2 = a_n Q_n (a_n Q_n - 2P_n),$$

Q_n also divides $D - P_{n+1}^2$. Therefore

$$Q_{n+1} = \frac{D - P_{n+1}^2}{Q_n} \in \mathbb{Z}.$$

$Q_{n+1} \neq 0$, because D is not a square.

Finally,

$$D - P_{n+1}^2 = Q_n Q_{n+1} \quad \Longrightarrow \quad Q_{n+1} \mid D - P_{n+1}^2.$$

The same hypothesis is ready for the next step.

Step 2: can these integers keep growing?

We now have, for every n,

$$x_n = \frac{P_n + \sqrt{D}}{Q_n}, \qquad P_n, Q_n \in \mathbb{Z}, \quad Q_n \neq 0.$$

There are still infinitely many possible integer pairs.

Track the other root as well:

$$x_n' = \frac{P_n - \sqrt{D}}{Q_n}.$$

We will first locate x_n', then use x_n and x_n' together to bound P_n and Q_n.

The conjugate follows the same algebraic step

The arithmetic we proved gives

$$x'_{n+1} = \left(\frac{1}{x_n - a_n}\right)' = \frac{1}{(x_n - a_n)'}$$
$$= \frac{1}{x'_n - a_n}.$$

The digit in both formulas is $a_n = \lfloor x_n \rfloor$.

For $x_0 = \sqrt{3}$, we use $a_0 = 1$, so

$$x'_1 = \frac{1}{-\sqrt{3} - 1} = \frac{1 - \sqrt{3}}{2}.$$

Here $\left\lfloor -\sqrt{3} \right\rfloor = -2$: taking a floor does not commute with conjugation.

Express a tail using the original number

Let $p_k/q_k = [a_0; \ldots, a_k]$. Lecture 1's prefix matrix gives, for $n \geq 2$,

$$x = \frac{p_{n-1}x_n + p_{n-2}}{q_{n-1}x_n + q_{n-2}}.$$

Cross-multiply and collect the terms containing x_n:

$$(q_{n-1}x - p_{n-1})x_n = p_{n-2} - q_{n-2}x.$$

Thus

$$x_n = -\frac{q_{n-2}x - p_{n-2}}{q_{n-1}x - p_{n-1}}.$$

Why the conjugates eventually become negative

Since $p_k/q_k \to x$ and $x' \neq x$, for all large k,

$$\left|\frac{p_k}{q_k} - x\right| < \frac{|x' - x|}{2}.$$

Thus $x' - p_k/q_k$ has the same sign as $x' - x$.

Conjugate the tail formula and factor out the positive q's:

$$x'_n = -\frac{q_{n-2}\left(x' - p_{n-2}/q_{n-2}\right)}{q_{n-1}\left(x' - p_{n-1}/q_{n-1}\right)}.$$

For large n, numerator and denominator have the same sign.

$$x'_n < 0 \quad \text{for all sufficiently large } n.$$

One more step gives a fixed interval

Once $x'_n < 0$ with $n \geq 1$, we have $a_n \geq 1$, so

$$x'_n - a_n < -1.$$

Taking the reciprocal gives

$$-1 < x'_{n+1} = \frac{1}{x'_n - a_n} < 0.$$

The same argument applies at every later step.

Therefore all sufficiently late tails satisfy

$$x_n > 1, \qquad -1 < x'_n < 0.$$

Bound the denominator first

For all sufficiently large n,

$$x_n = \frac{P_n + \sqrt{D}}{Q_n} > 1, \qquad -1 < x_n' = \frac{P_n - \sqrt{D}}{Q_n} < 0.$$

Subtract the two expressions:

$$x_n - x_n' = \frac{2\sqrt{D}}{Q_n} > 1.$$

The numerator is positive, so first $Q_n > 0$.

Multiplying by this positive denominator then gives

$$0 < Q_n < 2\sqrt{D}.$$

Now bound the numerator integer

Use $Q_n > 0$ in the conjugate inequalities:

$$-1 < \frac{P_n - \sqrt{D}}{Q_n} < 0 \quad \Longrightarrow \quad -Q_n < P_n - \sqrt{D} < 0.$$

Therefore

$$\sqrt{D} - Q_n < P_n < \sqrt{D}.$$

Since $Q_n < 2\sqrt{D}$,

$$P_n > \sqrt{D} - Q_n > -\sqrt{D}.$$

Together,

$$|P_n| < \sqrt{D}, \qquad 0 < Q_n < 2\sqrt{D}.$$

Step 3: only finitely many pairs remain

Fix D. Every sufficiently late pair lies in

$$|P_n| < \sqrt{D}, \qquad 0 < Q_n < 2\sqrt{D}, \qquad P_n, Q_n \in \mathbb{Z}.$$

Each integer has only finitely many choices.

For example, with $D = 3$ the possible bounds are

$$P_n \in \{-1, 0, 1\}, \qquad Q_n \in \{1, 2, 3\}.$$

There are infinitely many tails but only finitely many such pairs.

$$(P_i, Q_i) = (P_j, Q_j) \quad \text{for some } i < j.$$

A repeated pair forces repeating digits

Equal pairs give $x_i = x_j$. Hence

$$a_i = \lfloor x_i \rfloor = \lfloor x_j \rfloor = a_j, \qquad x_{i+1} = \frac{1}{x_i - a_i} = \frac{1}{x_j - a_j} = x_{j+1}.$$

Repeat the argument:

$$x_{i+k} = x_{j+k}, \qquad a_{i+k} = a_{j+k} \quad (k \geq 0).$$

With $\ell = j - i$, the expansion is

$$x = [a_0; \ldots, a_{i-1}, \overline{a_i, \ldots, a_{j-1}}].$$

This proves the converse of Lagrange's theorem.

Your turn: find the period of $\sqrt{6}$

Start with $D = 6$ and $(P_0, Q_0) = (0, 1)$.

$$a_n = \left\lfloor \frac{P_n + \sqrt{6}}{Q_n} \right\rfloor, \qquad 2 < \sqrt{6} < 3.$$

$$P_{n+1} = a_n Q_n - P_n, \qquad Q_{n+1} = \frac{6 - P_{n+1}^2}{Q_n}.$$

Record (P_n, Q_n, a_n) until an entire pair repeats.

Then write the continued fraction, marking where the period begins.

The calculation for $\sqrt{6}$

n	P_n	Q_n	x_n	a_n
0	0	1	$\sqrt{6}$	2
1	2	2	$(2+\sqrt{6})/2$	2
2	2	1	$2+\sqrt{6}$	4
3	2	2	$(2+\sqrt{6})/2$	2

$x_3 = x_1, \qquad \sqrt{6} = [2; \overline{2, 4}].$

The equal first two digits alone would not have proved repetition.

Lagrange's theorem is now complete

$$\text{eventually periodic CF} \iff \text{quadratic irrational.}$$

Lecture 1: a repeating tail gives a quadratic equation; a finite prefix preserves quadratic irrationality.

Today: integer pairs, eventual bounds, and a repeated tail.

We have proved that repetition must occur.

The remaining question is when it starts at the first digit.

Now ask for pure periodicity

A quadratic irrational x is called reduced if

$$x > 1, \qquad -1 < x' < 0.$$

We proved that all sufficiently late tails have these inequalities, and that the inequalities persist under the next step.

Galois's theorem

$$x \text{ is purely periodic} \iff x \text{ is reduced.}$$

For example, $\sqrt{3}$ is not reduced, but $(1+\sqrt{3})/2$ is.

Finiteness alone permits an initial segment

A function on the finite set $\{0, 1\}$ can satisfy

$$F(0) = 1, \qquad F(1) = 1.$$

Starting at 0 gives

$$0,\ 1,\ 1,\ 1, \ldots$$

Finiteness forces a repeated state, but the start need not return.

To remove the initial segment for a reduced surd, we will show:

A tail has at most one reduced predecessor.

Recover the previous digit from the conjugate

Suppose a reduced u has digit $a = \lfloor u \rfloor$ and next tail z:

$$z = \frac{1}{u - a}, \qquad z' = \frac{1}{u' - a}.$$

Solve the second equation for $-1/z'$:

$$-\frac{1}{z'} = a - u'.$$

Since $-1 < u' < 0$, we have $0 < -u' < 1$, and therefore

$$a < -\frac{1}{z'} < a + 1.$$

Thus the digit and the predecessor are determined by z:

$$a = \lfloor -1/z' \rfloor, \qquad u = a + \frac{1}{z}.$$

Rewind the repetition to the start

If x_0 is reduced, every x_n is reduced.

The finite-pair proof gives $x_i = x_j$ for some $i < j$.

$$
\begin{aligned}
x_i &= x_j \\
&\Downarrow \quad \text{unique reduced predecessors} \\
x_{i-1} &= x_{j-1} \\
&\vdots \\
x_0 &= x_{j-i}.
\end{aligned}
$$

The deterministic algorithm now repeats from its first digit.

A pure period must also be reduced

Suppose $x = [\overline{a_0, \ldots, a_{\ell-1}}]$.

Lecture 1 proves that x is quadratic.

Removing whole periods gives

$$x_{k\ell} = x_0 = x \qquad (k = 1, 2, \ldots).$$

We proved that every sufficiently late tail satisfies

$$x_n > 1, \qquad -1 < x_n' < 0.$$

Choose k large enough. Since $x_{k\ell} = x$, these hold for x.

This completes both directions of Galois's theorem.

The distinction in two examples

All quadratic surds are eventually periodic. For pure periodicity, check the two inequalities:

$$x > 1, \qquad -1 < x' < 0.$$

x	x'	Purely periodic?
$\sqrt{2}$	$-\sqrt{2} < -1$	No
$1+\sqrt{2}$	$-1 < 1-\sqrt{2} < 0$	Yes

$$\sqrt{2} = [1;\overline{2}], \qquad 1+\sqrt{2} = [\overline{2}].$$

Exit question: what does each step prove?

Lagrange: why did we need

$$\text{integer pairs} \quad + \quad \text{bounds?}$$

Galois: what extra fact let us turn

$$x_i = x_j \quad \text{into} \quad x_0 = x_{j-i}?$$

Explain why $\sqrt{3}$ has a period even though its conjugate is below -1.

Why the initial denominator can be negative

If the desired root is

$$x = \frac{-B - \sqrt{D}}{2A},$$

replace the equation by $-Ax^2 - Bx - C = 0$. Its plus-sign formula is

$$x = \frac{B + \sqrt{D}}{-2A}.$$

For example, $x = -\sqrt{2}$ satisfies $-x^2 + 2 = 0$, giving

$$D = 8, \qquad P_0 = 0, \qquad Q_0 = -2, \qquad x = \frac{0 + \sqrt{8}}{-2}.$$

Here $Q_0 \mid D - P_0^2$, since $-2 \mid 8$.

The integer pairs for $\sqrt{3}$

n	P_n	Q_n	x_n	a_n
0	0	1	$\sqrt{3}$	1
1	1	2	$(1+\sqrt{3})/2$	1
2	1	1	$1+\sqrt{3}$	2
3	1	2	$(1+\sqrt{3})/2$	1

$$\sqrt{3} = [1; \overline{1, 2}].$$

The initial tail is not reduced. The next two form a reduced cycle.

Your turn: check both inequalities

Which of these have purely periodic continued fractions?

$$\sqrt{2}, \qquad 1+\sqrt{3}, \qquad \frac{1+\sqrt{5}}{2}, \qquad \frac{\sqrt{2}}{2}$$

Check both inequalities:

$$x > 1, \qquad -1 < x' < 0.$$

Which of the four are eventually periodic?

The four tests, checked

x	x'	Purely periodic?
$\sqrt{2}$	$-\sqrt{2}$	No
$1+\sqrt{3}$	$1-\sqrt{3}$	Yes
$(1+\sqrt{5})/2$	$(1-\sqrt{5})/2$	Yes
$\sqrt{2}/2$	$-\sqrt{2}/2$	No

All four are eventually periodic.

A visible prefix can belong to the period

$$[2;\overline{1,2}] = 1+\sqrt{3} = [\overline{2,1}].$$

Its conjugate is $1-\sqrt{3} \in (-1,0)$.

Galois detects pure periodicity independently of where we first place the bar.

The conjugate reverses the period

If $x = [\overline{b_0, \ldots, b_{\ell-1}}]$, then

$$-\frac{1}{x'} = [\overline{b_{\ell-1}, \ldots, b_0}].$$

Each digit matrix is symmetric, so reversing a product transposes it.

$$N = \begin{pmatrix} A & B \\ C & D_0 \end{pmatrix}: \quad Cx^2 + (D_0 - A)x - B = 0.$$

The conjugate satisfies this equation too. Substitute $x' = -1/z$:

$$Bz^2 + (D_0 - A)z - C = 0,$$

the fixed-point equation for N^T.

The same matrix produces Pell solutions

Lecture 1 gave

$$U = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}, \qquad T_U(\sqrt{3}) = \sqrt{3}.$$

Ordinary matrix multiplication gives

$$U \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 2p + 3q \\ p + 2q \end{pmatrix}.$$

$$(2p + 3q)^2 - 3(p + 2q)^2 = p^2 - 3q^2.$$

Starting from $(1, 0)$ produces

$$(2, 1),\ (7, 4),\ (26, 15), \ldots,$$

each satisfying $p^2 - 3q^2 = 1$.